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NIM : 251524048

Kelas : 2B-D4

LATIHAN 1 (RegEx → String)

1. P → x(y)+z P = {xyz, xyyz, xyyyyz, ...}
2. Q → m | nr* Q = {m, n, nr, nrrr, ...}
3. R → (p | q)*r R = {r, pr, qqr, pqpr, ...}
4. T → t*(u | v) T = {u, v, tu, ttv, ...}
5. W → x(y | z)+ W = {xy, xz, xyz, xzzy, ...}

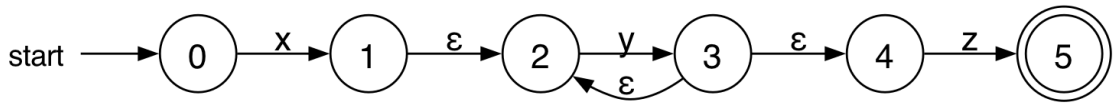
LATIHAN 2 (Membuat RegEx)

1. ^[A-Z]+$
2. ^[A-Za-z0-9]+$
3. ^[0-9]+$
4. ^[a-z0-9_]+@[a-z]+\.com$
5. ^[[:alnum:][:punct:]]+\.(jpg|png|gif)$

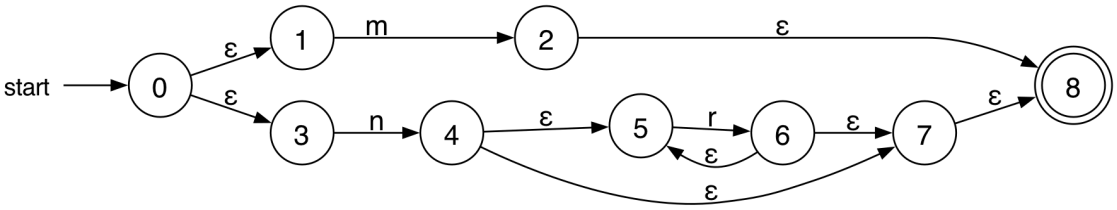
(baris 18 sampai 35 dibiarkan kosong, cukup foto bagian atas lembar ini)

LATIHAN 4 (RegEx → NFA)

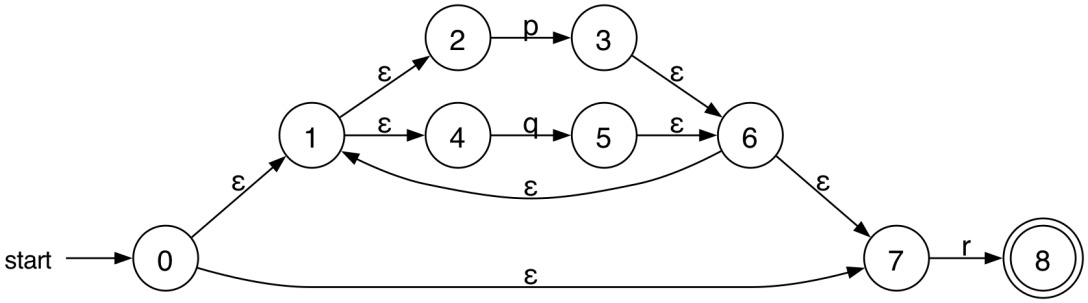
1) P → x(y)+z



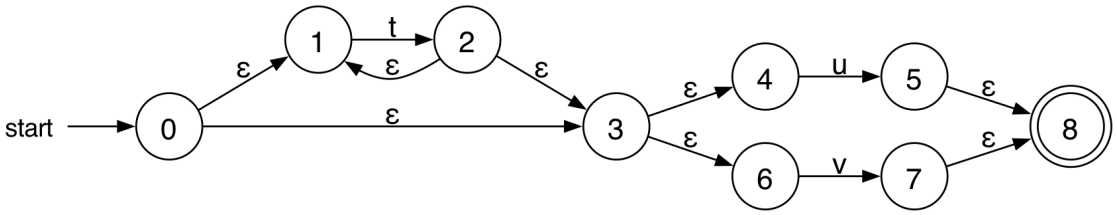
2) Q → m | nr*



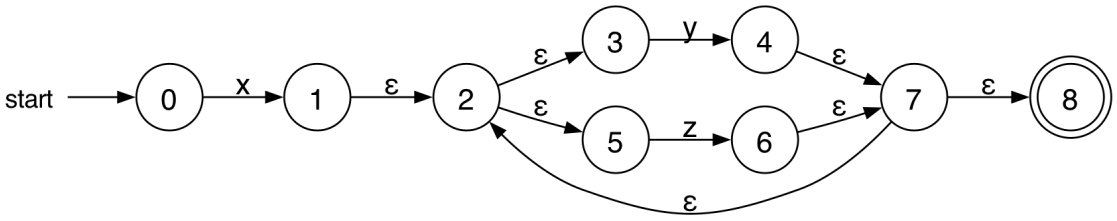
3) R → (p | q)*r



4) T → t*(u | v)



5) W → x(y | z)+



LATIHAN 5 (NFA → DFA) ε-cl = ε-closure, mv = move, * = state final

1) P → x(y)+z

ε-cl(0) = {0} → A

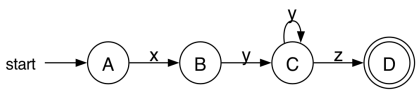
mv(A,x) = {1}, ε-cl = {1,2} → B

mv(B,y) = {3}, ε-cl = {2,3,4} → C

mv(C,y) = {3} → C

mv(C,z) = {5}, ε-cl = {5} → D

	x	y	z
→A	B	∅	∅
B	∅	C	∅
C	∅	C	D
*D	∅	∅	∅



2) Q → m | nr*

ε-cl(0) = {0,1,3} → A

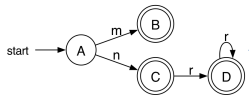
mv(A,m) = {2}, ε-cl = {2,8} → B

mv(A,n) = {4}, ε-cl = {4,5,7,8} → C

mv(C,r) = {6}, ε-cl = {5,6,7,8} → D

mv(D,r) = {6} → D

	m	n	r
→A	B	C	∅
*B	∅	∅	∅
*C	∅	∅	D
*D	∅	∅	D



3) R → (p | q)*r

ε-cl(0) = {0,1,2,4,7} → A

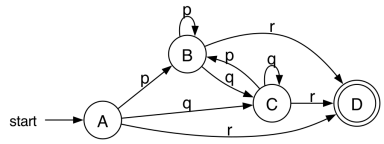
mv(A,p) = {3}, ε-cl = {1,2,3,4,6,7} → B

mv(A,q) = {5}, ε-cl = {1,2,4,5,6,7} → C

mv(A,r) = {8}, ε-cl = {8} → D

B, C: p → {3} = B, q → {5} = C, r → {8} = D

	p	q	r
→A	B	C	D
B	B	C	D
C	B	C	D
*D	∅	∅	∅



4) T → t*(u | v)

ε-cl(0) = {0,1,3,4,6} → A

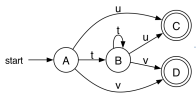
mv(A,t) = {2}, ε-cl = {1,2,3,4,6} → B

mv(A,u) = {5}, ε-cl = {5,8} → C

mv(A,v) = {7}, ε-cl = {7,8} → D

B: t → {2} = B, u → {5} = C, v → {7} = D

	t	u	v
→A	B	C	D
B	B	C	D
*C	∅	∅	∅
*D	∅	∅	∅



5) W → x(y | z)+

ε-cl(0) = {0} → A

mv(A,x) = {1}, ε-cl = {1,2,3,5} → B

mv(B,y) = {4}, ε-cl = {2,3,4,5,7,8} → C

mv(B,z) = {6}, ε-cl = {2,3,5,6,7,8} → D

C, D: y → {4} = C, z → {6} = D

	x	y	z
→A	B	∅	∅
B	∅	C	D
*C	∅	C	D
*D	∅	C	D

